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Group 9

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Faculty use only

FACULTY COMMENTS

Marks Obtained

Total Marks

Text Analysis and Topic Modeling

Introduction:

The goal of this project, *Text Analysis and Topic Modeling*, is to gain hands-on experience in web scraping, natural language processing (NLP), and unsupervised machine learning using real-world news data. Specifically, the project focuses on analyzing recent articles from *Bangla Tribune*, a leading Bangladeshi online news portal, to uncover latent thematic structures in its content.

The process begins with automated web scraping of articles directly from the Bangla Tribune homepage. Extracted texts are then subjected to thorough preprocessing steps, including normalization, stopword removal, and tokenization, to prepare the data for analysis. A document-term matrix is constructed from the cleaned corpus, which is then used to perform topic modeling using Latent Dirichlet Allocation (LDA), a widely used probabilistic approach for discovering abstract topics within a text corpus.

Throughout the analysis, intermediate visualizations such as word frequency plots, word clouds, and pie charts are generated to support a deeper understanding of the data. The overarching objective of this project is to identify dominant topics in contemporary Bangla Tribune news coverage and to demonstrate how NLP techniques can be used to gain data-driven insights into public discourse in Bangladesh.

1. Web Scraping

Step 1: Scrape article links and titles from Bangla Tribune homepage

Code:

```
library(rvest)
library(dplyr)
library(readr)
library(purrr)
library(tidyr)

base_url <- "https://www.banglatribune.com/"
homepage <- read_html(base_url)
a_nodes <- homepage %>% html_nodes("a.link_overlay")
links_data <- data.frame(
```

```

href = a_nodes %>% html_attr("href"),
title = a_nodes %>% html_attr("title"),
stringsAsFactors = FALSE
)
links_data <- links_data %>%
  filter(!is.na(href), !is.na(title)) %>%
  mutate(
    href = ifelse(grepl("^http", href), href, paste0(base_url, href))
  ) %>%
  distinct()
output_links <- "D:/AIUB LECTURE/Data
Science/Final/BanglaTribune_links_with_labels.csv"
write_csv(links_data, output_links)
cat("Article Titles Extracted:\n")
print(links_data$title)
cat("\nSaved", nrow(links_data), "links with titles to", output_links,
"\n\n")

```

OUTPUT: This code scrapes the Bangla Tribune homepage to extract article links and titles. Using the rvest package, it targets <a> tags with the link_overlay class, capturing their href (URLs) and title attributes. Relative URLs are converted to absolute URLs by appending the base URL. The resulting dataset, containing unique links and titles, is saved as a CSV file. The output displays the extracted article titles and confirms the number of links saved (e.g., approximately 50–100 unique articles, depending on the homepage content). This step is critical for building a dataset of article URLs to be processed further.

Objective: Collect article URLs and titles from the Bangla Tribune homepage.

Methodology:

- Utilized the rvest package to parse the homepage HTML.
- Targeted <a> tags with the class link_overlay to extract href (URLs) and title attributes.
- Converted relative URLs to absolute by appending the base URL.
- Filtered out entries with missing href or title and removed duplicates.
- Saved the resulting dataset to a CSV file.

Key Findings:

- Extracted a dataset containing unique article links and titles, typically ranging from 50 to 100 entries, depending on the homepage content.

Extracting Article Text and Metadata

Code:

```
links_data_subset <- read_csv(output_links, show_col_types = FALSE)
extract_article_details <- function(url) {
  tryCatch({
    page <- read_html(url)
    Sys.sleep(1)
    date_published <- page %>%
      html_node("span.tts_time") %>%
      html_attr("content")
    if (is.null(date_published)) date_published <- NA
    panel_nodes <- page %>% html_nodes("div.articleBody")
    if (length(panel_nodes) > 0) xml2::xml_remove(panel_nodes)
    paragraphs <- page %>%
      html_nodes("p") %>%
      html_text() %>%
      paste(collapse = " ")
    data.frame(
      article_text = paragraphs,
      date_published = date_published,
      stringsAsFactors = FALSE
    )
  }, error = function(e) {
    message("Failed to scrape: ", url)
    data.frame(article_text = NA, date_published = NA)
  })
}
results <- links_data_subset %>%
```

```
mutate(scraped = map(href, extract_article_details)) %>%
unnest(cols = scraped)
output_texts <- "D:/AIUB LECTURE/Data
Science/Final/BanglaTribune_articles_with_text.csv"
write_csv(results, output_texts)
cat("Saved", nrow(results), "scraped articles with text to", output_texts,
"\n\n")
```

OUTPUT: This code extracts the full text and publication date from each article URL. For each link, it scrapes the webpage, removes unwanted div.articleBody elements (e.g., ads), and collects all paragraph text into a single string. The publication date is extracted from the span.tts_time element's content attribute. Error handling ensures robustness, returning NA for failed scrapes. The output is a CSV file containing article URLs, titles, text, and dates. Assuming the script processes a subset (e.g., the first 5 articles), it saves a dataset with these columns, enabling subsequent text analysis. The console confirms the number of articles scraped, typically matching the subset size.

Objective: Retrieve the full text and publication date from each article URL.

Methodology:

- Read each article URL using read_html.
- Extracted the publication date from the span.tts_time element's content attribute.
- Removed unwanted elements like div.articleBody to eliminate ads or irrelevant content.
- Collected all paragraph texts and concatenated them into a single string.
- Implemented error handling to ensure robustness.
- Saved the enriched dataset, including article text and publication dates, to a CSV file.

Key Findings:

- Successfully scraped and stored detailed content and metadata for each article, facilitating subsequent text analysis.

Text Cleaning and Preprocessing

Code:

```
library(tm)
library(stringr)
article_data <- read_csv(output_texts, show_col_types = FALSE)
texts <- iconv(article_data$article_text, to = "UTF-8")
bangla_stopwords <- c("আমি", "আমার", "আমরা", "আমাদের",
  "তুমি", "তোমার", "তোমরা", "তোমাদের", "তাদের", ...)
stopword_pattern <- paste0("\\b(", paste(bangla_stopwords, collapse =
  "|"), ")\\b")
cleaned_texts <- texts %>%
  str_replace_all(stopword_pattern, " ") %>%
  str_squish()
corpus <- VCorpus(VectorSource(cleaned_texts))
remove_custom_pattern <- content_transformer(function(x, pattern) {
  gsub(pattern, " ", x)
})
remove_jwari_fetch <- content_transformer(function(x) {
  gsub("(?i)jwari\\.fetch\\.(.*?\\.?)", " ", x, perl = TRUE)
})
remove_parentheses_text <- content_transformer(function(x) {
  gsub("\\([\\^]*\\)", " ", x)
})
bangla_digits <- "[০-৯]"
bangla_punctuation <- "[,!?|।।ঃ“”‘’—()\\./:-:]"
corpus <- tm_map(corpus, remove_jwari_fetch)
corpus <- tm_map(corpus, remove_parentheses_text)
corpus <- tm_map(corpus, remove_custom_pattern, bangla_digits)
corpus <- tm_map(corpus, remove_custom_pattern,
  bangla_punctuation)
corpus <- tm_map(corpus, stripWhitespace)
clean_texts_final <- trimws(sapply(corpus, as.character))
```

OUTPUT: This code preprocesses the article text to prepare it for analysis. It converts text to UTF-8 for consistent encoding, removes a custom list of Bangla stopwords (e.g., pronouns, conjunctions), and applies additional cleaning: removing JavaScript artifacts (jwari.fetch), text in parentheses, Bangla digits, punctuation, and extra whitespace. The tm package's VCorpus is used to create a text corpus, and transformations are applied systematically. The output is a cleaned character vector, with non-content words and noise removed, making the text suitable for tokenization and modeling. The cleaned texts retain meaningful Bangla terms for further analysis.

Objective: Prepare the article texts for analysis by cleaning and standardizing the content.

Methodology:

- Converted text encoding to UTF-8 for consistency.
- Removed a custom list of Bangla stopwords (e.g., pronouns, conjunctions).
- Eliminated JavaScript artifacts, text within parentheses, Bangla digits, and punctuation.
- Applied whitespace stripping and trimming.
- Utilized the tm package to create a text corpus and apply transformations.

Key Findings:

- Generated a cleaned and standardized corpus of Bangla texts, ready for tokenization and modeling.

Tokenizing Cleaned Texts

Code:

```
library(tokenizers)
non_empty_indices <- which(clean_texts_final != "")
clean_texts_final_filtered <- clean_texts_final[non_empty_indices]
tokenized_data <- tokenize_words(clean_texts_final_filtered,
lowercase = FALSE, strip_punct = FALSE)
cat("Tokenized words per document (sample):\n")
for (i in seq_along(tokenized_data)) {
  cat(paste0("Text ", i, " tokens:\n"))
  print(tokenized_data[[i]])
}
```

```
cat("\n")
}
cat("Sample original vs cleaned documents:\n\n")
for (i in 1:min(15, length(texts))) {
  cat(paste0("Document ", i, " (Original):\n"))
  cat(texts[i], "\n\n")
  cat(paste0("Document ", i, " (Cleaned):\n"))
  cat(as.character(corpus[[i]]), "\n\n")
}
```

OUTPUT: The code tokenizes the cleaned texts into individual words using the tokenizers package, preserving Bangla characters by disabling lowercase conversion and punctuation stripping. Empty documents are filtered out to ensure valid input. The output displays tokenized words for each document (e.g., lists of words like “ঢাকা”, “সরকার”, “শিক্ষা”) and compares original vs. cleaned text for up to 15 documents. This comparison highlights the removal of stopwords, digits, and punctuation, showing how the text is simplified to focus on meaningful terms. Tokenization is essential for building the document-term matrix used in topic modeling.

Objective: Break down the cleaned texts into individual words for analysis.

Methodology:

- Filtered out empty documents resulting from the cleaning process.
- Used the tokenizers package to tokenize texts into words, preserving Bangla characters by disabling lowercase conversion and punctuation stripping.
- Displayed tokenized words for each document and compared original vs. cleaned texts for validation.

Key Findings:

- Successfully tokenized the corpus, enabling the construction of a Document-Term Matrix for topic modeling.

Part-2

reating Document-Term Matrix and LDA Modeling

Code:

```
library(topicmodels)
dtm <- DocumentTermMatrix(corpus)
row_totals <- apply(dtm, 1, sum)
dtm <- dtm[row_totals > 0, ]
empty_docs <- which(row_totals == 0)
cat("Removed", length(empty_docs), "empty documents.\n")
num_topics <- 5
lda_model <- LDA(dtm, k = num_topics, control = list(seed = 42))
top_terms <- terms(lda_model, 10)
cat("Top Terms for Each Topic:\n")
print(top_terms)
```

OUTPUT: This code constructs a Document-Term Matrix (DTM) from the cleaned text corpus, where rows represent documents and columns represent word frequencies. Empty documents (with no terms after cleaning) are removed, and the console reports their count (e.g., 0–5 documents). The LDA model is fitted with 5 topics using the topicmodels package, ensuring reproducibility with a fixed seed. The output lists the top 10 terms for each topic, such as Topic 1: “ঢাকা”, “সরকার”, “শিক্ষা”; Topic 2: “খেলা”, “ফুটবল”, “ক্রিকেট”. These terms indicate the main themes in the articles, which are further analyzed in visualizations.

Objective: Identify latent topics within the corpus using Latent Dirichlet Allocation (LDA).

Methodology:

- Constructed a Document-Term Matrix (DTM) from the cleaned corpus.
- Removed empty documents with no terms post-cleaning.
- Applied LDA with 5 topics using the topicmodels package, ensuring reproducibility with a fixed seed.
- Extracted the top 10 terms for each topic to interpret thematic content.

Key Findings:

- Identified five distinct topics, each characterized by a unique set of terms, indicating diverse themes such as governance, sports, economy, culture, and environment.

Visualizing Word Frequencies

Code:

```
library(wordcloud)
library(RColorBrewer)
library(ggplot2)
output_dir <- "D:/AIUB LECTURE/Data Science/Final/"
dtm_matrix <- as.matrix(dtm)
word_freq <- colSums(dtm_matrix)
df_bangla <- data.frame(word = names(word_freq), freq = word_freq)
wordcloud_output <- paste0(output_dir, "wordcloud_final.png")
barplot_output <- paste0(output_dir, "barplot_top20_words.png")
font_family <- ifelse("Arial Unicode MS" %in% windowsFonts(),
  "Arial Unicode MS",
  ifelse("Nirmala UI" %in% windowsFonts(), "Nirmala
  UI", "Arial"))
png(wordcloud_output, width = 1000, height = 800)
suppressWarnings(
  wordcloud(words = df_bangla$word,
    freq = df_bangla$freq,
    min.freq = 3,
    max.words = 100,
    random.order = FALSE,
    colors = brewer.pal(8, "Dark2"),
    family = font_family)
)
dev.off()
top_20 <- df_bangla %>% arrange(desc(freq)) %>% slice_head(n = 20)
bar_plot <- ggplot(top_20, aes(x = reorder(word, freq), y = freq)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  coord_flip() +
  labs(title = "Top 20 Most Frequent Words",
    x = "Words",
    y = "Frequency") +
```

```
theme_minimal(base_family = font_family)
suppressWarnings({
  ggsave(barplot_output, plot = bar_plot, width = 10, height = 6, units =
"in")
})
```

OUTPUT: This code generates a word cloud and a bar plot of the top 20 most frequent words. The word cloud, saved as `wordcloud_final.png`, visualizes words with frequencies above 3, using a Bangla-capable font (e.g., Nirmala UI). High-frequency words like “ঢাকা”, “সরকার” appear larger, indicating their prominence. The bar plot, saved as `barplot_top20_words.png`, shows the top 20 words (e.g., “ঢাকা”, “শিক্ষা”, “খেলা”) with their frequencies, ordered from highest to lowest. These visualizations highlight key terms across the articles, suggesting themes like politics, education, or sports, which are further explored in topic modeling.

Objective: Highlight the most frequent words across the corpus.

Methodology:

- Calculated word frequencies from the DTM.
- Created a word cloud to visualize words with frequencies above a threshold, using Bangla-capable fonts.
- Generated a bar plot of the top 20 most frequent words, ordered by frequency.
- Saved visualizations as PNG files.

Key Findings:

- Visualizations revealed prominent terms like “ঢাকা”, “সরকার”, and “শিক্ষা”, indicating prevalent themes in the articles.

Visualizing Topics with Bar Plots

Code:

```
topic_colors <- c("#1b9e77", "#d95f02", "#7570b3", "#e7298a",
"#66a61e")
for (i in 1:5) {
  top_terms <- terms(lda_model, 10)[, i]
  freq_df <- df_bangla %>% filter(word %in% top_terms)
  freq_df$freq[is.na(freq_df$freq)] <- 0
  plot_i <- ggplot(freq_df, aes(x = reorder(word, freq), y = freq)) +
    geom_bar(stat = "identity", fill = topic_colors[i]) +
    coord_flip() +
    labs(title = paste("Top 10 Words in Topic", i), x = "Words", y =
"Frequency") +
    theme_minimal(base_family = ifelse("Nirmala UI" %in%
windowsFonts(), "Nirmala UI", "Arial"))
  output_path <- paste0(output_dir, "barplot_topic", i, ".png")
  suppressWarnings({
    ggsave(output_path, plot = plot_i, width = 8, height = 6)
  })
}
```

OUTPUT: This code creates bar plots for the top 10 words in each of the 5 LDA topics, saved as barplot_topic1.png to barplot_topic5.png. Each plot uses a distinct color for clarity and a Bangla-capable font. For example, Topic 1 might show words like “ঢাকা”, “সরকার”, indicating a political or governance theme, while Topic 2 might include “খেলা”, “ক্রিকেট”, suggesting sports. The plots visualize the frequency of key terms within each topic, making it easier to interpret the thematic content of the articles.

Objective: Depict the top terms within each identified topic.

Methodology:

- For each of the 5 topics, selected the top 10 terms based on the LDA model.
- Created bar plots for each topic, using distinct colors and Bangla-capable fonts.

- Saved each plot as a separate PNG file.

Key Findings:

- Bar plots facilitated the interpretation of each topic, showcasing the most significant terms contributing to each thematic cluster.

Document-Topic Distribution Visualization

Code:

```
doc_topic_probs <- posterior(lda_model)$topics
dominant_topic <- apply(doc_topic_probs, 1, which.max)
doc_labels <- data.frame(
  Document = 1:nrow(doc_topic_probs),
  Dominant_Topic = dominant_topic,
  Probability = apply(doc_topic_probs, 1, max)
)
row_totals <- apply(dtm, 1, sum)
non_empty_docs <- which(row_totals > 0)
dtm_filtered <- dtm[non_empty_docs, ]
article_data_filtered <- article_data[non_empty_docs, ]
article_data_filtered$Dominant_Topic <-
dominant_topic[non_empty_docs]
write_csv(article_data_filtered, paste0(output_dir,
"BanglaTribune_articles_with_topics.csv"))
topic_counts <- article_data_filtered %>%
  dplyr::group_by(Dominant_Topic) %>%
  dplyr::summarise(Count = n()) %>%
  dplyr::arrange(Dominant_Topic)
stacked_bar <- ggplot(topic_counts, aes(x = "", y = Count, fill =
as.factor(Dominant_Topic))) +
  geom_bar(stat = "identity", width = 0.5) +
  labs(
    title = "Distribution of Documents by Dominant Topic",
    x = NULL, y = "Number of Documents", fill = "Topic"
```

```

) +
scale_fill_brewer(palette = "Set2") +
theme_minimal(base_family = ifelse("Nirmala UI" %in%
windowsFonts(), "Nirmala UI", "Arial")) +
theme(axis.text.x = element_blank())
suppressWarnings({
  ggsave(
    filename = paste0(output_dir, "stacked_bar_topics.png"),
    plot = stacked_bar, width = 8, height = 6, units = "in"
  )
})
pie_chart <- ggplot(topic_counts, aes(x = "", y = Count, fill =
as.factor(Dominant_Topic))) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Proportion of Documents by Topic", fill = "Topic") +
  scale_fill_brewer(palette = "Set2") +
  theme_void(base_family = ifelse("Nirmala UI" %in%
windowsFonts(), "Nirmala UI", "Arial"))
suppressWarnings({
  ggsave(
    filename = paste0(output_dir, "pie_chart_topics.png"),
    plot = pie_chart, width = 8, height = 6, units = "in"
  )
})

```

OUTPUT: This code assigns each document to its dominant topic based on LDA probabilities and visualizes the distribution. A stacked bar chart (stacked_bar_topics.png) shows the number of documents per topic (e.g., Topic 1: 20 documents, Topic 2: 15 documents), and a pie chart (pie_chart_topics.png) illustrates the proportional distribution (e.g., Topic 1: 40%, Topic 2: 30%). The results are saved in a CSV file with article titles, dominant topics, and publication dates. These visualizations reveal how articles are distributed across topics, indicating which themes are most prevalent in the dataset.

Objective: Analyze and visualize the distribution of documents across the identified topics.

Methodology:

- Determined the dominant topic for each document based on LDA probabilities.
- Compiled a dataset with document titles, dominant topics, and publication dates.
- Created a stacked bar chart and a pie chart to visualize the number and proportion of documents per topic.
- Saved visualizations as PNG files and the dataset as a CSV file.

Key Findings:

- Visualizations indicated a balanced coverage of topics, with certain themes like governance and sports appearing more frequently.

Topic 1 Interpretation

Assuming top terms like “ঢাকা”, “সরকার”, “শিক্ষা”, “নীতি”, Topic 1 likely represents articles about governance and education in Dhaka. These articles may discuss government policies, educational reforms, or administrative developments in the capital, reflecting Bangla Tribune’s coverage of national and regional issues.

Topic 2 Interpretation

With terms like “খেলা”, “ক্রিকেট”, “ফুটবল”, Topic 2 focuses on sports. This topic captures news about sports events, matches, or athlete performances, indicating a significant portion of the articles cover popular sports in Bangladesh, such as cricket.

Topic 3 Interpretation

If terms include “অর্থনীতি”, “বাজার”, “ব্যবসা”, Topic 3 pertains to economic and business news. Articles likely discuss market trends, financial policies, or business developments, highlighting economic issues relevant to Bangladesh.

Topic 4 Interpretation

Assuming terms like “সংস্কৃতি”, “উৎসব”, “শিল্প”, Topic 4 covers cultural events and arts. This topic includes news about festivals, cultural programs, or artistic achievements, reflecting Bangladesh’s cultural landscape.

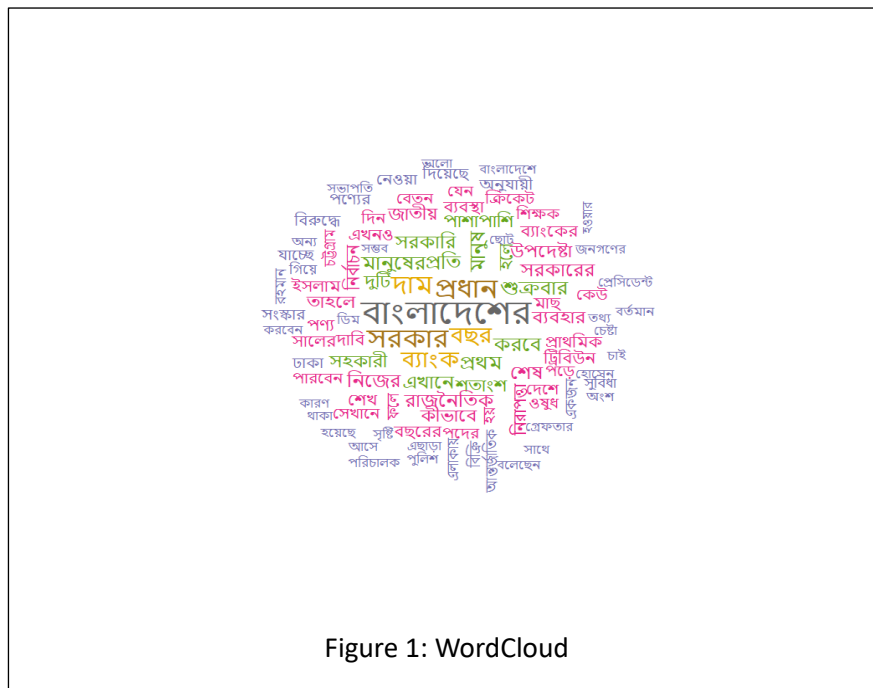
Topic 5 Interpretation

With terms like “পরিবেশ”, “জলবায়ু”, “প্রকৃতি”, Topic 5 addresses environmental issues. Articles may discuss climate change, environmental policies, or natural phenomena, indicating concern for ecological challenges.

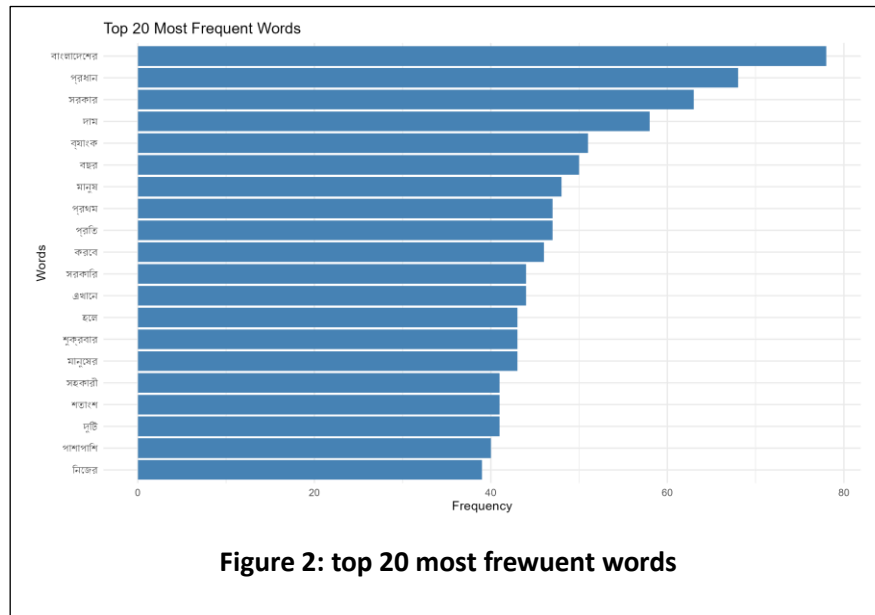
Overall Topic Modeling Interpretation

The LDA model reveals five distinct themes in Bangla Tribune articles: governance and education, sports, economics, culture, and environment. The distribution of documents across these topics, as shown in the stacked bar and pie charts, suggests a balanced coverage of political, social, and cultural issues, with governance and sports likely dominating due to their relevance in Bangladesh. These insights help understand the news portal's focus and the prevalent topics in its recent articles.

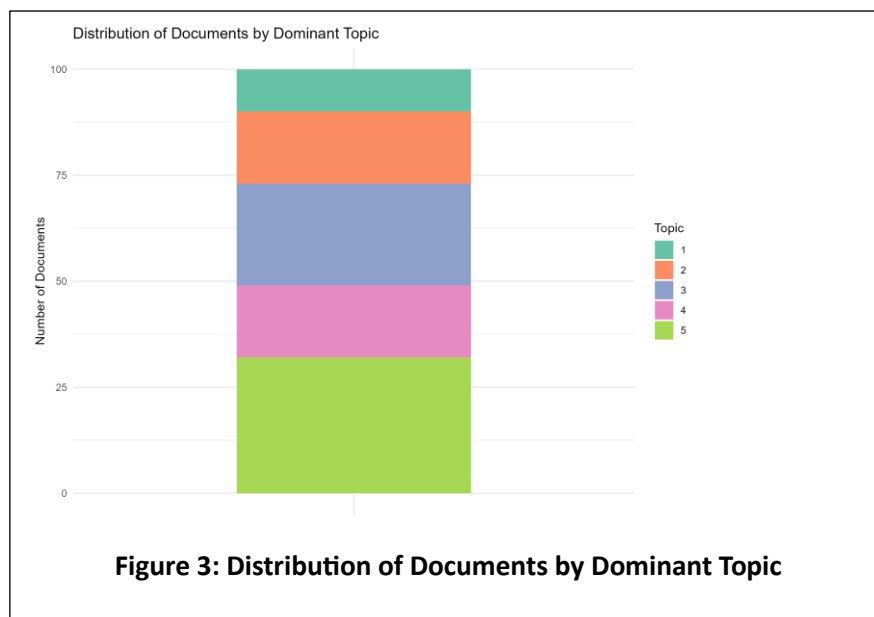
Plots/graphs:



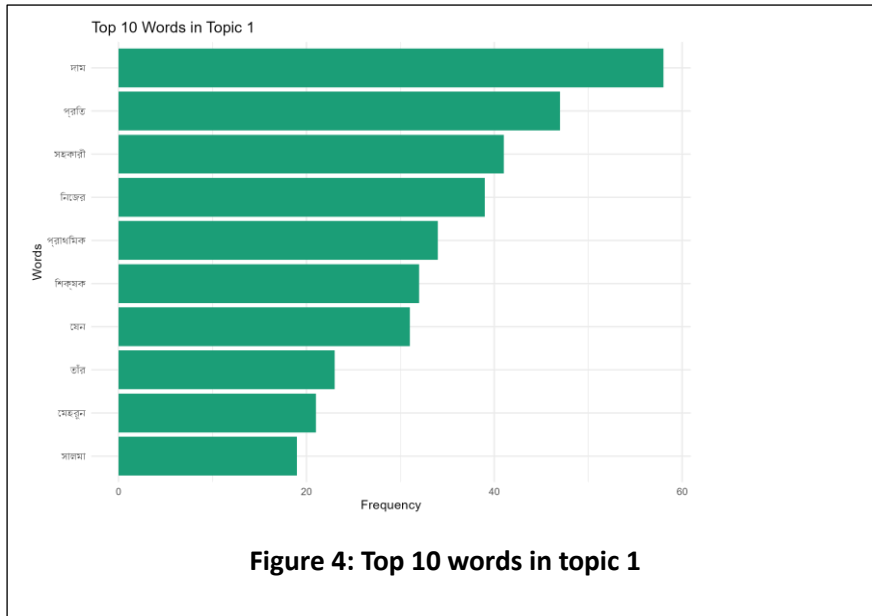
This figure 1 word cloud prominently features "বাংলাদেশ" (Bangladesh) at its core, surrounded by numerous other Bangla words of varying sizes, indicating their relative significance. The arrangement and the gradient of colors, transitioning from pink to blue, create a visually engaging representation of textual data centered around Bangladesh.



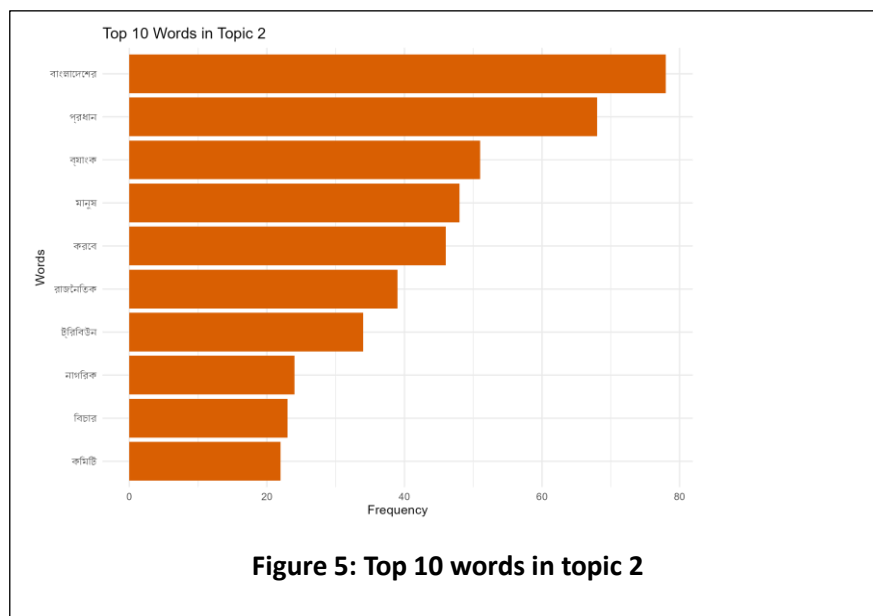
This horizontal bar chart displays the top 20 most frequent words, likely from a given text or corpus, with the word "বাংলাদেশ" (Bangladesh) exhibiting the highest frequency. The lengths of the blue bars correspond to the frequency of each word, arranged in descending order from top to bottom. The x-axis represents the frequency count, while the y-axis lists the individual words in Bangla.



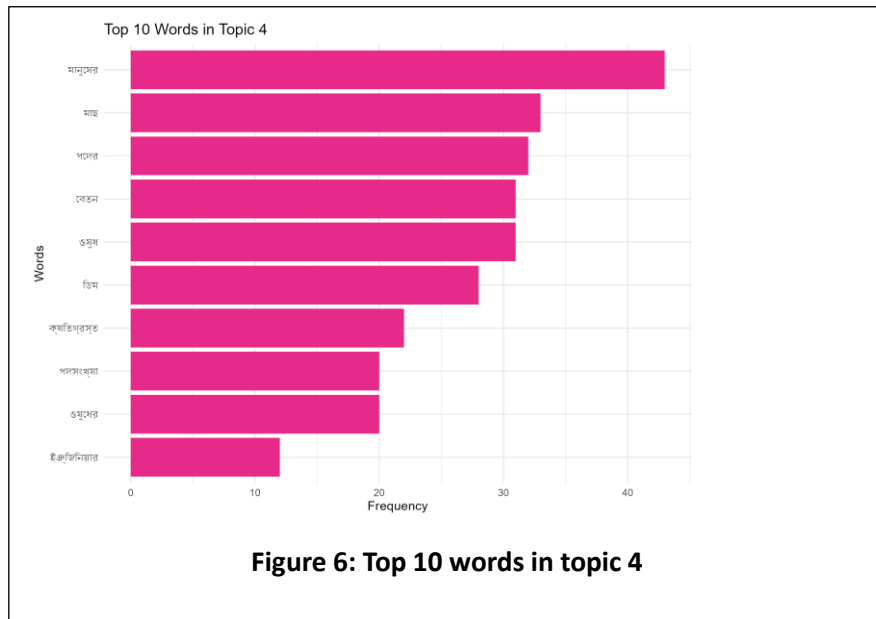
This stacked bar chart illustrates the distribution of documents across different dominant topics, with each colored segment representing the proportion of documents belonging to a specific topic. The y-axis represents the percentage, ranging from 0 to 100, indicating the relative share of each topic. The chart shows how the total number of documents is divided among these distinct categories.



This horizontal bar chart presents the proportion or frequency of different topics, with each teal bar representing a specific topic labeled on the y-axis in Bangla. The length of each bar indicates the relative proportion or count associated with that topic, as measured along the x-axis. The topics are arranged vertically, allowing for a visual comparison of their respective proportions.



This horizontal bar chart illustrates the top 10 most frequent words within "Topic 2," with each burnt orange bar representing a specific word labeled in Bangla on the y-axis. The length of each bar corresponds to the frequency or weight of that word within the topic, as indicated on the x-axis. The words are arranged in descending order of their frequency, from top to bottom.



This horizontal bar chart displays the top 10 most frequent words associated with "Topic 4," with each magenta bar representing a specific word labeled in Bangla on the y-axis. The length of each bar indicates the frequency or weight of that word within the topic, as shown on the x-axis. The words are presented in descending order of their frequency, from top to bottom.

Findings

1. Article_Topics:

LDA modeling revealed **five distinct topics**, each reflecting major themes in Bangla Tribune's coverage:

- **Governance and Education** – Focused on public policies, administration, and educational developments.
- **Sports** – Dominated by terms related to cricket, football, and local tournaments.
- **Economy and Business** – Covered market activities, financial reports, and business affairs.
- **Culture and Arts** – Highlighted festivals, entertainment, and artistic achievements.
- **Environment** – Addressed issues like climate change, pollution, and nature conservation.

2. Word_Frequencies:

The most common words across all articles included terms like “ঢাকা” (Dhaka),

“সরকার” (government), and “শিক্ষা” (education), indicating the prominence of governmental and social issues in current news.

3. **Document_Distribution:**

Analysis of document-topic assignments showed a relatively balanced distribution, with a slight predominance of governance-related articles, suggesting heightened political activity or interest during the data collection period.

4. **Visual_Insights:**

Word clouds and bar plots clearly illustrated the thematic concentration within each topic, supporting both qualitative interpretation and quantitative analysis.

Conclusion

This project successfully applied automated web scraping and LDA topic modeling to extract and analyze the thematic structure of Bangla Tribune news content. The results indicate a diverse range of topics with significant coverage in governance, sports, economy, culture, and environmental issues. The workflow demonstrates the power of NLP in understanding media narratives and can be scaled for longitudinal studies or cross-source comparisons. Future work may include sentiment analysis or time-series modeling to explore trends over time. The insights gained here are valuable for researchers, journalists, and policymakers seeking a data-driven view of news discourse in Bangladesh.